$$c_{j,t} = c_j^0 + u_{\theta,t}(c_j^0), \quad (F_{j,t})_{:,k} = e_k + \frac{u_{\theta,t}(c_j^0 + \ell_p e_k) - u_{\theta,t}(c_j^0 - \ell_p e_k)}{2\ell_p}$$